Holiday Light Co.

EST. · FIELD OPERATIONS

Field Training Manual

Department: Field Operations

Chapters: 12

Classification: Internal Use Only

Table of Contents

Part I

Getting Started

CHAPTER 1

Welcome

New Hire Paperwork & Onboarding

Before your first day in the field, the following must be completed:

New Hire Onboarding Checklist

- ☐ Employment Agreement
- ☐ HR Platform
- ☐ Team Chat
- ☐ Payroll Setup
- ☐ I-9
- ☐ Drivers License
- ☐ Account Setup & Access
- ☐ Project Photo App
- ☐ Gas Card
- ☐ W4
- ☐ State Tax Form

Seasonal Schedule Overview

BLACK OUT DAYS

Black Out Days are reserved schedule days — no one may call out or use vacation time on these dates.

The holiday lighting season follows a general timeline each year. Exact dates shift, but the pattern stays consistent.

Sep	Oct	Nov	Dec	Jan	Feb

Pre-Season · Ramp-Up · Peak Installs · Wind-Down · Removals

Two-Phase Contract Structure

In-Season Contract: Sept 26 – Dec 10 — Installation of holiday lighting projects. Mon–Fri (weekends as needed). 10-hour day at flat base pay.
Post-Season Contract: Jan 2 – Feb 1 — Takedown and removal of lighting installations. Mon–Fri (flexible as needed). 10-hour day at flat base pay.
Bonuses: Employees who complete each contract without termination receive a performance-based bonus. Post-Season pay may be adjusted based on In-Season performance.

Month	Phase	Schedule Details
September	Pre-Season	Weekdays: Black Out Days • Weekends: Off
October	Ramp-Up	Weekdays: Black Out Days • Saturdays: On Call • Sundays: Off
In-Season Contract (Sept 26 – Dec 10)		
November	Peak Installs	Nearly all days are Black Out Days • Commercial work begins early Nov • Thanksgiving + day after: Off
December	Wind-Down	Week 1: Black Out Days on weekdays • Week 2+: On Call shifts • Late Dec: Days off
Post-Season Contract (Jan 2 – Feb 1)		
January	Removals	Weekdays: Black Out Days through most of the month • Saturdays taper off
February	Overflow	All remaining removal work completed • Schedule flexes as needed

Team Structure

★ Field Operations Manager

- Oversees all field operations including scheduling, crew performance, job completion, and field logistics

Supervises: Installation Crews, Quality Control, Repair Crews

Senior & Junior Technicians

- Execute lighting installations with emphasis on safety, attention to detail, and quality
- Work in 2-man crews (Senior Tech + Junior Tech)

Reports to: Field Operations Manager

☑ Quality Control Rep

- Conducts post-installation checks to ensure jobs meet quality standards and client expectations
- Gathers customer feedback and supports crew accountability

Reports to: Field Operations Manager

⚡ Repair Technician

- Respond to service calls and perform repairs on damaged or malfunctioning lighting setups
- Ensure ongoing customer support during the season

Reports to: Field Operations Manager

Chapter 1 — Key Takeaways

✦Complete all onboarding paperwork before your first day in the field

✦In-Season Contract: Sept 26 – Dec 10 (installs) • Post-Season: Jan 2 – Feb 1 (takedowns)

✦November is peak season — nearly all days are Black Out Days

✦All crews work in 2-man teams and report to the Field Operations Manager

✦Quality Control Reps inspect every job to ensure quality standards and client expectations

Chapter 2

Daily Operations

Daily Schedule

Your day runs on a tight timeline. Arrive at 7:00 AM sharp.

Warning

Arriving after 7:15 AM will result in an attendance deduction and may be documented.

1

Arrival — 7:00–7:15 AM

Senior:

Pull up vehicle, review work order with Junior Tech.

Junior:

Morning checklist, clean vehicle, stage supplies for the day (C9's, minis, plugs, clips, ground stakes, spheres, specialty items).

2

Daily Prep — 7:15–7:45 AM

Senior:

Meet with Operations Manager — review repairs, discuss adjustments/training, routing, questions.

Junior:

Finish cleaning, stage work order supplies, load ladders, organize production aisle.

3

Departure — 7:45–8:00 AM

Senior:

Double-check staged items, pull specialty items, record morning checklist on CompanyCam, leave by 8 AM.

Junior:

Load approved items, final sweep of production aisle.

4

Return / End of Day — ~5–7 PM

Senior:

Return signed work orders, help offload, set RYOBI chargers on block, log end-of-day photo to CompanyCam, place key in lock box.

Junior:

Offload all items, place empty containers in proper location, consolidate supplies, stage end cap.

Workday Expectations

You are paid for a 10-hour workday at a flat daily rate; however, your day ends when your last project is finished.

Finished Early

You have 2 projects worth $3,000. You complete both by 3pm. Check in with your Operations Manager — once approved, you're free to clock out. You will still receive full-day pay.

Running Late

You have 2 projects worth $3,000. The jobs take longer than expected. You are expected to finish the projects before returning to the shop and clocking out.

Remember

Work Mon–Fri, weekends as needed during peak season. All assigned projects **must** be completed.

Conduct & Expectations

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Uniforms

White Shirt, Green Pants, Company Hat, and Belt

⛽

Expense Cards

Gas card per truck. Keep all receipts. Drop in key drop box daily

💬

Work Chats

All work chats will remain professional in conduct

Allowed

- ✅ Smoking with discretion (never on customer property)
- ✅ Music in headphones or speaker (mind lyrics)
- ✅ Respectful customer communication

Not Allowed

- ❌ Drugs & Alcohol — we reserve the right to drug test
- ❌ Unprofessional conduct in work chats
- ❌ Arguing with customers — escalate to your Ops Manager

Customer Escalation

If a customer has an issue you cannot resolve, keep your composure and contact your Operations Manager immediately.

Chapter 2 — Key Takeaways

- ✦Arrive by 7:00 AM — after 7:15 is an attendance dock
- ✦You are paid for a 10-hour day at flat rate; your day ends when your last project is done
- ✦Consistent tardiness may lead to a formal write-up or disciplinary action
- ✦Uniform, receipts, clean vehicle, and professional conduct are non-negotiable

Chapter 3

Software & Apps

Three apps power your daily workflow. Learn them well — you will use them every day.

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Payroll Platform

Pay info, timesheets, time off, benefits, and employment documents

📅 JUL 17

Field Service App

View daily schedule, job details, and close out completed work

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Project Photo App

Upload job photos, complete checklists, scan documents, view nearby projects

Project Status Tags

These tags are applied to projects in the Photo App to track their current state:

Tag	Meaning
In Progress	
Awaiting Additional Product	Job needs more materials before it can be completed
Partially Complete	Some but not all line items are finished
Needs Boom to Complete	Job requires a boom lift to finish
Needs Garland/Wreath	Job is waiting on greenery materials
Change Orders	
Change Order	A change order has been submitted
Change Order Complete	Change order work is finished
Quality & Closeout	
Failed Quality Assurance	Job did not pass QA inspection
Invoice Sent	Invoice has been sent to customer
Uncontrollable Damage	Damage from weather or other uncontrollable factors

Chapter 3 — Key Takeaways

- ✦The Payroll Platform manages your pay, timesheets, and time-off requests
- ✦The Field Service App is where you view schedules and close out jobs
- ✦The Project Photo App documents every install — always complete checklists
- ✦Learn the status tags so you can communicate project state accurately

Part II

Tools & Materials

CHAPTER 4

Tools & Equipment

Know your tools inside and out. Every installer carries these daily.

Little Giant Ladder

A-frame with adjustable lock mechanisms

Extension Ladder

Propping up, raising with rope, ladder spotting

Glue Gun

Load glue & batteries. Clean surfaces before applying

Staple Gun

Loading & techniques. Never staple through wire

Flathead Screwdriver

Prying open flashing and prying out nails

Needle Nose Pliers

Cutting lines & wire, closing plugs

Measuring Wheel

Roofline & ground measurements. Add 10% extra for slack and connections

Bailing Wire & Zip Ties

Greenery & railing installs. Cut zip tie tails

Painters Pole & Limb Cutters

Hanging/removing lights on hedges. Trim overgrown branches

PRO TIP

Clear out cracks and clean the surface before applying glue for the strongest hold.

DANGER

Never cut live wire. Stapling through the line will cause a power short.

TRAINING VIDEOS

See training videos for Little Giant setup, Extension Ladder use, and all tool techniques.

Chapter 4 — Key Takeaways

- ✦Always carry pliers, staple gun, screwdriver, and tester bulb on your person
- ✦Add 2 extra feet when measuring — short lines cause rework
- ✦Clean surfaces before gluing; never staple through wire
- ✦Cut zip tie tails after fastening — loose tails look sloppy

CHAPTER 5

Vehicle Use & Checklists

Vehicle Expectations

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Cleanliness

Remove trash daily. Keep tools and materials organized in designated locations

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Gas Receipts

Drop receipts in key drop box EOD. Log miles, cost, and date

🚗

Safe Driving

Drive with traffic flow. Know your exits and turns in advance

Vehicle Expenses Tracking Sheet

Track the following daily. Use one row per day; total all columns at the end of the month.

Day	Total Miles	Gas Gal.	Gas Cost	Mileage	Service	Tolls	Repairs	Wash	Misc
1–31	…	…	…	…	…	…	…	…	…

Fleet Truck Morning Check-List

Complete this checklist every morning before leaving the shop.

🔧 Tool Checklist

- ☐ Tool Box (1)
- ☐ Staple Gun (2)
- ☐ Electric Glue Gun (1)
- ☐ Big Battery — Glue Gun (1)
- ☐ Small Battery — Glue Gun (1)
- ☐ Corded Glue Gun (1)
- ☐ Battery Charger (1)
- ☐ Needle Nose Pliers (2)
- ☐ Flathead Screwdrivers (2)
- ☐ Measuring Wheel (1)
- ☐ Knife / Box Cutter (1)
- ☐ Mini Spool Bailing Wire (1)
- ☐ Wrench (1)
- ☐ Windex (1)
- ☐ Paper Towels (1)
- ☐ 22ft A-frame Ladder (1)
- ☐ Additional Ladder 18–28ft (1)

📦 Product Checklist

- ☐ Bucket of "Warm White" C9 (1)
- ☐ Bucket of "Red" C9 (1)
- ☐ Bucket of Clips (1)
- ☐ Bucket of Stakes (1)
- ☐ Spool of Socket Wire (2)
- ☐ Spool of Blank Wire (2)
- ☐ Bin with Minis (2)
- ☐ Container: Male/Female Plugs (1)
- ☐ Container: Staples, Glue, Zip Ties (1)

Chapter 5 — Key Takeaways

- ✦Run the morning checklist every day — missing tools means lost time on-site
- ✦Keep the vehicle clean — a tidy truck reflects professionalism
- ✦Drop gas receipts in the key drop box every night
- ✦Track daily vehicle expenses and total them monthly

CHAPTER 6

Materials

Understanding your materials is just as important as knowing your tools. This chapter covers every product you will use in the field.

Material	Details	Notes
LIGHTING		
Mini Lights	Trees, Railings, Garland, Wreaths, Pillars	Wrap balls; tie plugs together
C5 / C7 Bulbs	Gutters, Windows, Pathways, Roofline, Ridges	Match bulb size to application
Lighting Spheres	6″ and 9″ sizes	Handle carefully — fragile
CLIPS & ATTACHMENT		
Flashing Clips	Metal flashing / drip edge	Most common clip type
Magnet Clips	Magnetic surfaces	Strong hold, easy removal
Circle Clips	Rounded or tile surfaces	Also works on flashing
WIRE & CONNECTORS		
Socket Wire	Male or female plugs on either side	12″ or 15″ spacing
Blank Wire / Lamp Cord	Draws power to lighting products	No sockets
Vampire Plugs	Male vs Female • In-Line T's	Ribbed right • Cut wire clean
ACCESSORIES		
Ground Stakes	Line walkways & driveways	Add 5 ft to measurements
Photo-cell Timer	Powers on at sunset, 8-hour default	Avoid multiple timers per property

Attachment Methods

Surface	Method	Watch Out
Flashing	Flashing clips or magnet clips	—
Tiles	Circle clips or flashing clips	—
Wood Eaves	Staples	Staples must NOT go through wire
Glue-safe surfaces	Hot glue	NEVER glue on stucco

Danger

Never glue on stucco. Staples must never go through the wire — this causes a power short.

Caution — Timers

Avoid using multiple timers if possible. Each outlet receives different light levels, causing lights to turn on at different times. If multiple timers are necessary, split them logically (e.g., one for the home, one for landscaping).

Quick Reference: Plug Orientation

✦ MALE PLUGS ✦

Male plugs always point toward the power source. **Female plugs** always point away from power. When a step says "Female Out" or "Female @ the top," it means orient the female end so it faces outward or sits at the top, ready to receive the next connection back toward power.

Chapter 6 — Key Takeaways

- Match the clip type to the surface — flashing clips, magnet clips, or circle clips
- Never glue on stucco; never staple through wire
- Add 5 extra feet when measuring for ground stakes
- Male plugs always point toward power; female plugs point away
- Avoid multiple timers per property — they cause uneven on-times

Part III

Installation Techniques

CHAPTER 7

Ladder Safety & Elevated Work

Ladder safety and elevated work platform protocols are non-negotiable. Know the right setup for every situation.

SAFETY FIRST

Always have a spotter when using extension ladders. Use counter weight on tall saguaros. Never rush ladder setup.

Ladder Safety Rules

1

Pre-Use Inspection

Before using any ladder, complete three checks: pre-start inspection, function/setup check, and workplace/surroundings check. If any component is damaged or a safety label is unreadable,

tag it out of service

and remove it from use.

2

The 4:1 Setup Rule

For extension ladders: place the base

1 foot out for every 4 feet up

. This is the correct setup angle for stability.

3

3-Point Contact

Always face the ladder, keep your belt buckle between the rails, and maintain

three points of contact

at all times (two hands + one foot, or two feet + one hand).

4

Roof Edge Transition

When stepping off at a roof edge, rails must

extend at least 3 feet above the landing

and be secured if required.

Do

- ✅ Use non-conductive (fiberglass) ladders near electrical sources
- ✅ Fully open stepladders and lock spreaders before use
- ✅ Always have a ground person stabilize the ladder
- ✅ Tag out damaged ladders immediately

Don’t

- ❌ Never stand on the top step or top cap of a stepladder
- ❌ Never move or “walk” a ladder while someone is on it
- ❌ Never use metal ladders near energized electrical sources
- ❌ Never skip rungs or carry heavy items in both hands while climbing

Ladder Types

Little Giant Ladder

Multipurpose: A-frame, extension, scaffold, and staircase configurations. *(see training video)*

Extension Ladder

Use the 4:1 rule. Raise with rope, always have a spotter. *(see training video)*

Ladder Use by Surface

Surface	Setup	Key Rule
Palm Trees	Two man job. Ladders on opposite sides	Start from top, wrap down. Female plug at top
Standard Trees	A-frame for limbs and trunk	Never lean extension against branches
Cactus / Saguaro	A-frame if possible. One man front, one back	**Always** use counter weight on tall ones. Ladders must mirror
Canopies	Lean into thicker branches. Tie rope around branch	Spotter at bottom of ladder at all times. Never climb onto the canopy

Training Videos

See training videos for Werner Ladder Safety, Little Giant setup, extension ladder technique, and ladder spotting.

Fall Protection & Harness Use

When working from MEWPs (boom lifts, aerial lifts), fall protection equipment is mandatory. No exceptions.

MANDATORY

You **must** wear a full-body harness every time you operate or ride in a boom lift.

1. **Pre-Use Harness Inspection**
   Before each use, inspect for
   frayed webbing, damaged buckles, and worn stitching
   . If any damage is found, remove it from service immediately and report it.

2. **Attach to Dorsal D-Ring**
   The lanyard must attach to the
   dorsal (back) D-ring
   — not the side or front D-ring. This is the correct fall-arrest attachment point.

3. **Anchor Inside the Basket**
   When in a boom lift, your lanyard must be attached to the
   designated anchor point inside the basket
   — never wrapped around the railing.

PROHIBITED

Never stand on the mid-rail or top rail of a MEWP basket for extra reach. Never operate if the platform is not level or stable — lower the platform, stop work, and report the issue.

MEWP Operations (Boom Lifts & Aerial Lifts)

*MEWP stands for **Mobile Elevated Work Platform**. Only trained and authorized personnel may operate a MEWP.*

1. **Pre-Operation Inspection**
   Complete a full pre-operation inspection before every use. Check controls, hydraulics, tires, and safety devices.

2. **Check for Overhead Hazards**
   Always check for overhead hazards (power lines, branches, structures) before raising the platform.

3. **Operate & Work**
   Maximum occupancy is determined by the
   manufacturer's rated capacity
   . Never exceed it. Do not operate on slopes without stabilizers.

4

Report Defects

Report any MEWP defects or malfunctions before the next use. Use ground controls for emergency lowering if needed.

10-Foot Rule

When operating a boom lift near **power lines**, you must maintain a minimum distance of **10 feet** at all times.

Hard Hats

Hard hats are required when working below or near MEWP operations. Fall protection is required in scissor lifts even though they have guardrails.

Chapter 7 — Key Takeaways

- 4:1 rule for extension ladders — 1 foot out for every 4 feet up
- Maintain 3-point contact at all times; face the ladder, belt buckle between rails
- Tag out any damaged ladder immediately — do not use it
- Harness is mandatory on all boom lifts — attach lanyard to dorsal (back) D-ring
- Maintain 10 feet minimum from power lines when operating MEWPs
- Complete pre-operation inspection on every MEWP before use
- Palm trees are always a two-man job; use counter weight on tall saguaros

Chapter 8

Landscape Lighting

Landscape lighting transforms a property. Follow these procedures for professional results every time.

Wiring at Ground Level

1. **Plan Your Route**
   Be efficient. Wire should always run behind trees and bushes in discreet locations.

2. **Cross Walkways Safely**
   If wire crosses a walkway or step, hot glue it into the crack so it is not a tripping hazard.

3. **Waterproof Grass Trees**
   Trees in grass must be waterproofed — electrical tape connections up to 4 feet off the ground.

Starting Trees

1. **Determine Starting Point**
   Standard trees start at the bottom. Palms always start at the top underneath the bulk of the palm.

2. **Tie Plugs Together**
   Always tie plugs together — keeps consistent spacing, simplifies takedowns, and reduces repairs. *(see training video)*

3. **Fasten & Staple**
   Fasten plug on the backside. Staple plug to tree — always staple behind the mini driver.

4. **Wrap the Middle First**
   Prioritize the middle of the tree. Wrap the middle first and shoot off from there.

Standard Trees

1. **Start on the Back Side**
   Begin away from street view. If the tree has multiple trunks, starting points should meet in the middle.

2

Wrap Up the Center Trunk

Use 5 finger spacing until you get to the first limb.

3

"Double Back" on Limbs

Continue wrapping the main trunk after finishing small branches. Only double-back on small branches, not full limbs. Twigs can just be "outlined."

4

Check from Street View

Step back and look from the street to ensure limbs and trunks look finished. *(see training video)*

❄ 5 FINGER SPACING ❄

5 finger spacing — spread your fingers wide and use that hand-width as the gap between each wrap of lights. This keeps spacing consistent without needing a measuring tool.

Canopy Wrap

1

Connect Power

Plug into trunk lights or run a separate power source if the trunk is not wrapped.

2

Anchor on Back Side

Run minis to the back of the canopy and tie around a branch to anchor.

3

Wrap Consistently

About 2 feet between each line. Front-facing side: level, parallel lines. Back side: slanted to scale up.

4

Reach the Top

Always

go to the very top. Use a painters pole for tall trees while your partner holds slack. *(see training video)*

> **WARNING**
> An unwrapped canopy top is highly visible and will be noted during quality inspection.

Lighting Spheres

1. **Access the Canopy**
   See Chapter 7 (Ladder Safety) for canopy access techniques.

2. **Start High, Work Down**
   Begin at the highest point. Space spheres evenly for a full lighting effect.

3. **Call Ops Manager**
   Always
   call your Operations Manager when a sphere job is completed. *(see training video)*

Palm Trees

1. **Test Lights First**
   You won't have power until finished since you start from the top.

2. **Start from the Top**
   Wrap "Female Out" (female plug facing outward). Staple the line at the top to hold.

3. **Wrap Down**
   Use 5 finger spacing. Keep lines consistent and parallel.

4. **Finish at the Bottom**
   Staple the male plug on the backside of the palm tree.

Bushes & Hedges

Use the same wiring process as tree lighting. Run power to bushes before installing mini lights. Tie off around a branch on the bottom. The most important aspect is ensuring even light coverage.

1. **Drape Evenly**
   Drape lights evenly across the top surface of the bush or hedge.

2. **Push Into Foliage**
   Push lights slightly into the foliage for depth — avoid sitting only on the surface.

3. **Use Painter's Pole**
   For taller hedges, use a painter's pole with hook to guide and space the wire evenly.

4

Even Spacing

Maintain consistent spacing for a uniform glow. Avoid bunching at corners.

Troubleshooting Landscape

When lights aren't working, follow this diagnostic flow:

1. Check Outlet
2. Reset GFI
3. Check Breaker
4. Test with Bulb
5. Try New Ball

LIGHTS SHUT OFF RANDOMLY?

Most likely the lights got stapled or cut somewhere. Reset the GFI and plug in trees one at a time. Once the problem tree is found, unplug individual strands until you isolate the issue.

CHAPTER 8 — KEY TAKEAWAYS

- Wire runs behind trees in discreet locations — hot glue across walkways
- Always tie plugs together for consistent spacing and easier takedowns
- 5 finger spacing is the standard for tree wraps and palm wraps
- Always wrap to the very top of canopies — gaps are highly visible
- For bushes and hedges: drape evenly, push into foliage for depth, avoid bunching at corners
- Troubleshoot power issues systematically: outlet → GFI → breaker → test bulb → new ball

Chapter 9

Residential Roofline Installs

Roofline installs are the core of the holiday lighting season. This chapter covers the complete workflow from arrival to closeout.

Install Roles

Installation Crews

- Execute the full install — measuring, cutting, hanging, wiring, bulb check, cleanup, and job closeout

2-man crews are the most efficient grouping

Field Operations Manager

- Answers job questions, approves early clock-outs, handles escalations

Line item questions may go to the Sales rep

☑ Quality Control Rep

- Conducts post-installation checks to ensure quality standards and client expectations

Repair Technician

- Respond to service calls and perform repairs on damaged or malfunctioning setups

Crew Goals

2-man crews work best when goals are aligned. Discuss season goals and daily targets with your partner — having long-term and daily goals will drive your success as a crew.

Step-by-Step Workflow

1. **Contact Customer En Route**
   Send text or call when heading to the job.

2. **Arrive & Game Plan**
   View the work order and game plan with your partner.

3. **Split & Prep**
   One installer contacts the customer, the other preps — find power sources, prep wire, ladders, C9's, clips, minis.

4

Execute the Install

Follow your game plan. Rooflines: measure and cut. Trees: one runs wire, the other preps balls.

Always

keep tools on your person (pliers, staple gun, screwdriver, tester bulb).

5

Secure & Hide Wiring

Tuck wire on homes, glue wire through cracks. All wiring should be as hidden as possible.

6

Bulb Check

One installer on ground, one on roof. Walk each roofline together: every bulb on, secured, no slack, no corners jumped.

7

Job Walk Through

Review every line item. Pick up any trash on ground or roof.

8

Pack Up & Notify

Ensure all tools are back in the vehicle. Notify the homeowner of completion.

9

Close Out & Move On

Close the job on the Field Service App. Contact your next appointment.

Establishing Starting Points

1

Find Power Close By

Starting point should be near an outlet on either side of the home. A sloppy wire in the middle looks unprofessional and will need to be redone.

2

Test the Outlet

Use a tester bulb to confirm the outlet is active.

3

Determine Hanging Method

Clips on flashing, clips on tile, clips on gutters, staples, hot glue, or ground stakes.

> **DRIP EDGE**
>
> The drip edge is the metal flashing along the roofline where clips are inserted. Understanding the profile helps you choose the right clip type and insertion angle.

Measuring Rooflines

Measuring Wheel (Most Accurate)

- ✅ Always reset the dial before measuring
- ✅ Best: walk the roofline from on top of the roof
- ✅ For tile roofs: follow from ground level

Counting Tiles (Quickest)

- ❌ Not always the most accurate
- ❌ Curved tiles: peak to valley ≈ 1 foot
- ❌ Flat tiles: count each tile as 1 foot

Measurement Rule

Always **add 10% extra** to your measurement for slack and connections. Measure each section individually and account for corners, peaks, and returns.

2-Man Cut and Bulb

❧ KNUCKLE DOWN ❧

Knuckle Down — the "knuckle" is the built-in clip on C9 sockets. When prepping a line, always attach the multi-purpose clip with the same orientation on each socket — make sure the knuckle and socket point downward. This ensures every clip along the line has a consistent orientation.

1. **Measure & Cut**
   Installer 1 measures lines. Installer 2 pulls and cuts socket wire. Text measurements sequentially or keep a shared note.

2. **Keep Lines in Order**
   Set cut lines on the ground in the same order they were measured. Do not mix up the order.

3. **Bulb & Clip**
   One installer inserts bulbs, the other installs multi-purpose clips to each socket — knuckle down (built-in clip and socket pointing downward on every socket for consistent orientation). Finished lines go back in order on the ground.

4. **Send Lines to Roof**
   Installer 2 takes the first two lines up, setting rolled lines on their corresponding rooflines. Installer 1 throws up remaining lines sequentially.

Prepping Lines

1. **Unroll on Corresponding Line**
   Match each prepped line to its measured roofline section.

2. **Check Clip Direction**
   Always make sure clips face the right direction — clips never lie.

3. **Orient to Definite End**
   Unroll with the
   definite end
   in mind.

✦ DEFINITE END ✦

Definite End — a starting point or endpoint that will need an exact fit; no more roof line past that point.

PRO TIPS — PREPPING

Inside corners: Hook the line to something on the roof or leave part rolled at a stopping point.
Peaks: Find the middle bulb and place it at the peak.
Mixed surfaces: Some rooflines mix flashing, tile, gutters, and staples. “Flip the clips” as needed.
Efficiency: Installer 1 can pre-cut power lines while lines are being prepped.

Hanging Runs

RULE

Always start with definite ends facing street view. First bulb goes right up against the definite end.

CLIP SPACING

Place clips **every 12 inches** on shingles. Use the proper clip type for each surface. Maintain consistent bulb spacing and ensure clean lines — no sagging.

From the Roof (Fastest)

- ✅ Getting on the edge is necessary
- ✅ Always keep three points of contact

From a Ladder (Alternate)

- ❌ Move the ladder as little as possible
- ❌ Work at arm's length on both sides
- ❌ Use the "leap frog" method with partner

Tile Gables & Tight Flashing

Tile Gables: Use your screwdriver to loosen the nail, insert the clip with bulb, then hammer the nail back in with the butt end.
Tight Flashing: Wedge your screwdriver underneath to pry open.

Wiring Roof Line Lights

Each line connects with "zip cord" wire (blank wire with male and female plugs). All males point to power, females point away. Wire should always be tucked and hidden in corners between rooflines. Prep wire from the ground by counting tiles, or bring wire and plugs up to the roof.

Bulb & Quality Check

Quality Check — Walk-Through Items

- ☐ Every bulb is on
- ☐ Every bulb is secured
- ☐ Lines have no slack
- ☐ No corners jumped
- ☐ No loose minis
- ☐ No loose power cords

Installer 1 on the ground, Installer 2 on the roof. Walk each roofline together. Found errors are fixed immediately by Installer 2. If lights won't come on, troubleshoot or call your Operations Manager.

Securing Peaks & Cleanup

1. **Secure All Peaks**
   Put a #48 staple into the wood eave. No wooden eave? Use zip ties or bailing wire to hook the bulb to a nail through tile.

2. **Clear the Roof**
   No tools, clips, pliers, or wire left behind.

3 Ground Cleanup & Notify

Installer 1 collects ground trash and contacts customer. Check off line items on the work order and close out the job.

Chapter 9 — Key Takeaways

- ✦Follow the 9-step workflow for every roofline install — no shortcuts
- ✦Add 10% extra to every measurement for slack and connections
- ✦Clip placement every 12 inches — use proper clip type for each surface
- ✦Keep cut lines in order — mixing them up wastes time on the roof
- ✦Always start hanging from the definite end facing street view
- ✦Males point to power, females point away — always
- ✦Run the full quality checklist before leaving — every error costs points

CHAPTER 10

Windows, Walls, Stakes & Specialty

Specialty installations require attention to detail and surface awareness. Each surface type demands a different approach.

Windows

1. **Plan the Outline**
   A bulb in every corner, every "middle point," and one in between. Visualize bulb count when measuring.

2. **Mount to Surface**
   Most common: hot glue socket base to window face. Clean the glass first.

3. **Alternative Surfaces**
   Stone/brick: glue directly. Stucco:
   cannot
   be glued. Last resort: install eyelets at corners, string bailing wire, fasten bulbs to wire.

Surface Compatibility

SAFE TO GLUE

- ✅ Glass (clean first)
- ✅ Stone or brick (direct mount)
- ✅ Metal surfaces
- ✅ Wood eaves (or use staples)

CANNOT GLUE

- ❌ Stucco — never use hot glue
- ❌ If stucco walls require it, be very minimal
- ❌ Call Ops Manager for stucco arches

Walls & Ground Stakes

Feature	Method	Key Detail
Walls (Stone/Brick)	Hot glue	Bulbs point up. If not level, lay flat — keep in line
Walls (Stucco)	Minimal glue	Add 1 foot to measurement
Ground Stakes	Mallet or hammer	Add 4–5 extra feet. Uniform depth & inline

Railings, Pillars & Arches

Feature	Method	Key Detail
Railings (Minis)	Pass ball between railing bars	Consistent spacing is critical for quality
Railings (C9's)	Zip tie on either side of socket	Secure to top of railing
Pillars	Female plug at top, wrap down	6 finger spacing. Front side level, drops on back
Arches	Brick, stone, or metal only	Find middle bulb first, then hit each side. One glues, one holds

Stucco Arches

If a stucco surface requires an arch, **always** call your Operations Manager for approval before proceeding.

Chapter 10 — Key Takeaways

- Windows need a bulb in every corner and midpoint — uniform outlines only
- Never hot glue on stucco. Call Ops Manager for stucco approvals
- Ground stakes: add 4–5 extra feet and install uniform and inline
- Pillars use 6 finger spacing; arches start from the middle bulb

CHAPTER 11

Holiday Greenery

Greenery adds the finishing touch to holiday installations. Proper technique ensures they look full and stay secure.

Wreaths

6′	4′	3′
6-foot Wreath	**4-foot Wreath**	**3-foot Wreath**
Largest size. Statement piece for peaks	Standard peak or wall mount	Doors, gates, and fences

1. **Wrap with Mini Lights**
   Per the work order. Wreaths on peaks typically need hardware in the wood eave.

2. **Tile Peaks**
   Hook bailing wire to the nail at the peak and fasten the wire to the wreath's metal backing.

3. **Fences & Gates**
   Use bailing wire. For gates that open, use split wreaths — install each side separately and run power on both sides. *(see training video)*

Garland & Bows

1. **Match Garland Lengths**
   Always make sure garland lengths are equal on both sides.

2. **Secure to Frame**
   Install screws or staples to the door frame. Zip tie garland to screws or staples.

3. **Fluff & Finish**
   Make sure garland is fluffed. Bows go at the lowest point of the wreath or on wall sconces.

Chapter 11 — Key Takeaways

- ✦All wreaths need hardware — bailing wire for tile peaks, screws for wood eaves
- ✦Split wreaths for opening gates — power both sides independently
- ✦Garland lengths must be equal; always fluff before leaving
- ✦Bows go at the lowest point of wreaths or on wall sconces

Part IV

End of Season

CHAPTER 12

Removing Lights & Decor

Removals begin in early January once the holiday season wraps up. Removal work continues through January and into February as needed.

Removals Checklist

🚚 REMOVALS LOAD-OUT

- ☐ Empty 57gal Totes w/ Lids (12)
- ☐ Scrapers (2)
- ☐ Painters Poles (2)
- ☐ Limbcutter Extension (1)
- ☐ Pack of Notecards (1)
- ☐ Empty Clear Bucket (1)
- ☐ Pliers (2)
- ☐ Flathead Screwdrivers (2)
- ☐ Painters Hooks (2)
- ☐ Sharpies (2)
- ☐ Scotch Tape (1)

Removals Workflow

1. **Arrive & Contact Customer**
   Review work order, confirm address and scope. Notify the customer you're on-site.

2. **Walk the Property**
   Identify all installed lighting, decor, and connection points before starting.

3. **Disconnect Power & Record Video**
   Unplug all lighting circuits before beginning removal.
   Then film a walkthrough showing all power sources, wire routes, landscaping, walls, pillars, and fences.

4. **Remove Roofline Lighting**
   Start here. Ensure all clips are taken out of rooflines.

5 Remove Additional C9 Lighting

Take down walls, ground stakes, and any other C9 installations.

6 Remove Landscape Minis

Take down minis from trees, bushes, and hedges.

7 Inspect & Organize Materials

Check bulbs, clips, and wire for damage. Separate reusable from damaged. Roll each ball tightly and stage in front of the feature it belongs to.

8 Clean Up & Photo Document

Ensure no clips, wire, or debris remain. Take photos with balls staged in front of their assigned objects. Add text: strand count + color (e.g., “4 Red”).

Chapter 12 — Key Takeaways

- **Disconnect all power** before beginning any removal work
- Walk the property first to identify all lighting, decor, and connection points
- Remove roofline first, then C9s, then landscape minis
- Inspect and separate reusable materials from damaged; roll tightly and stage for photos
- Leave the property cleaner than you found it — no clips, wire, or debris left behind

APPENDIX

Reference

Glossary

Terms & Definitions

Term	Definition
Scheduling & Roles	
Black Out Day	A reserved schedule day — no one may call out or use vacation time on these dates
Quality Control Rep	Conducts post-installation checks to ensure jobs meet quality standards and client expectations. Also gathers customer feedback and supports crew accountability
Installation Techniques	
Knuckle Down	The “knuckle” is the built-in clip on C9 sockets. When prepping a line, attach the multi-purpose clip with the knuckle and socket pointing downward on every socket to ensure consistent clip orientation along the entire line
5 Finger Spacing	Spread your fingers wide and use that hand-width as the gap between each wrap. Keeps spacing consistent without a measuring tool
6 Finger Spacing	Same technique as 5 finger spacing but with slightly wider gaps — used on pillars
Definite End	A starting point or endpoint on a roofline that requires an exact fit — no more roof line past that point
Drip Edge	The metal flashing along a roofline where clips are inserted
Electrical & Wiring	
Male Plug	Always points toward the power source
Female Plug	Always points away from the power source. “Female Out” or “Female @ the top” means orient the female end outward or at the top
Zip Cord	Blank wire with a male and female plug on either side, used to connect lines of lights
Socket Wire	Wire with bulb sockets — male or female plugs can be installed on either side
Blank Wire / Lamp Cord	Wire used solely to draw power to lighting products (no sockets)
Vampire Plug	A plug that pierces through wire insulation to make an electrical connection. “Ribbed right” when wiring
In-Line T	A vampire plug configuration that creates a T-junction to split power to multiple runs
GFI	Ground Fault Interrupter — the reset button on outdoor outlets. First thing to check when power isn’t working
Products	
C5 / C7 / C9	Bulb sizes used for different applications — C9 is the most common for rooflines

Term	Definition
Photo-cell Timer	A light-activated timer that powers on at sunset. Standard setting is 8 hours

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